# YZV 405E - Natural Language Processing
# Project Guideline

**2025-2026 Spring**

*"The limits of my language mean the limits of my world."*

— *Ludwig Wittgenstein*

This semester, for your term project, you are given five different topics that touch upon various aspects of natural language processing and have been explored in different ways in the literature. Please choose the topic that interests you most from those described below.

For any questions, contact the lecture assistants (including **both** in the e-mail recipients) Doğukan Arslan (arslan.dogukan@itu.edu.tr) and Atakan Site (site21@itu.edu.tr).

## 1 Project Topics

Below, five project topics (three main topics, two of which include two subtasks each) are briefly explained. Please check provided links for further instructions.

### 1.1 AdMIRe 2

AdMIRe 2 (Advancing Multimodal Idiomaticity Representation) is a shared task that addresses the challenge of idiomatic language understanding by evaluating how well computational models interpret potentially idiomatic expressions (PIEs) using both text and images. Given a context sentence containing a PIE and a set of five images, the task is to rank the images based on how accurately they depict the meaning of the PIE used in that sentence. There are two subtasks in this challenge, and you may choose either one as your project topic. You need to work with every language listed for this task.

**Subtask 1: Image & Text** https://www.codabench.org/competitions/10547/

**Subtask 2: Text Only** https://www.codabench.org/competitions/10548/

### 1.2 PARSEME 2.0

PARSEME 2.0 is a shared task whose main objective is to identify and paraphrase multiword expressions (MWEs) in written text, aiming to deal with all syntactic types of MWEs. There are two subtasks: the first corresponds to the classical identification task in running text. The second consists in paraphrasing a sentence containing a MWE, so as to remove idiomaticity. You may choose either one as your project topic. You need to work with every language listed for this task.

**Subtask 1: MWEs identification** https://www.codabench.org/competitions/13186/

**Subtask 2: Paraphrasing MWEs** https://www.codabench.org/competitions/13192/

### 1.3 Tabular QA Dataset Translation

*** Students may only select this topic after obtaining approval from the instructors.**

This topic focuses on developing a novel machine translation pipeline for tabular data. You will work with an English table QA benchmark where each instance consists of a table paired with a natural language question and its answer. Your goal is to design a translation pipeline that produces a high-quality Turkish version of this benchmark. The translation should cover the main textual elements involved in the QA task, including (at least) questions, answers or answer spans, table headers, and any additional textual metadata that is necessary to understand and answer the questions.

Dataset: `https://drive.google.com/drive/folders/1mYJuqhSBNZuvIqpWXh-G9iyV-_WXyRxI`

**Recommended reading for this topic.** Students who are considering this topic and wish to better understand the problem are required to read the following papers before contacting us:

- **XINFOTABS**: Minhas et al. (2022), "XINFOTABS: Evaluating Multilingual Tabular Natural Language Inference". `https://aclanthology.org/2022.fever-1.7/`
- **Cross-Lingual Adaptation for Multilingual Table QA**: Cho et al. (2026), "Cross-Lingual Adaptation for Multilingual Table Question Answering and Comparative Evaluation with Large Language Models". `https://www.mdpi.com/2073-431X/15/2/92`
- **M3TQA**: Shu et al. (2025), "M3TQA: Massively Multilingual Multitask Table Question Answering". `https://arxiv.org/abs/2508.16265`

Students working on this topic are also required to design their own strategy for assessing translation quality at scale. This strategy should include at least one reference-free machine translation quality estimation approach. In their report, students must motivate their choice of metrics and analyse how the resulting quality scores relate to typical translation error types in tabular data, such as mistranslated headers, categorical values, numbers, dates or units. In addition, the translated data produced by each group will be manually inspected and graded by the course instructors.

## 2 Project Proposal

- You must form a team of **exactly** 3 people (except for Project 1.3). Please enter your project team information using the link **here**. There is also a section in the document for those who cannot find a teammate. Here, you can indicate your availability by entering your name and e-mail address, with the project topics you interested in. If you cannot find a teammate, contact the course assistants **before** the proposal submission deadline.
- Create a **private** GitHub repository[1] that you are expected to update **regularly** throughout the project. Name the repository as `YZV405E_2526_TeamName` and invite the course assistants (`@dgknrsln` and `@atakansite`) to the repository you have created. For everything else, follow widely accepted **best practices**.
- Enter the Codabench competition for your chosen topic. See Section 3 for details.
- You should **write a proposal** using the **ACL template**. The proposal should be **1 or 2 pages** long (except references and appendice). You should:
  - Use proposal title as: `[Team Name] at YZV405E: [Your Descriptive Title]`
  - Briefly introduce the problem and the dataset,
  - Have an **extensive** literature review (without merely listing papers),
  - Explain your candidate model**s** proposed to solve the problem, discussing why they are appropriate and how they operate.

[1] https://docs.github.com/en/repositories

- Your solution should be **deep learning** oriented, yet combining those methods with traditional one's is encouraged.
- The deadline for submitting project proposals is **March 3, 23:59**.

### 2.1 If your proposal is rejected..

- If your project proposal is rejected for any reason, you **must** update and re-upload it in line with the feedback you received.
- If your project proposal is rejected, you **will not** obtain any points from this part of the project, but you need to update it anyway in order to continue with the remaining parts.
- If your project proposal is rejected a second time, you **will not** be allowed to continue the project and **will fail** the course.
- You will have **1 week** after receiving feedback to upload your updated project proposal.

## 3 Codabench

Some term project topics have their own Codabench competition. For those:

- After signing up on Codabench, create an organization by clicking your username in the top-right corner, then selecting "Create Organization." Name it as your team name. Fill in the email field with one of the team member's emails, then submit. Click "Edit Organization" in the opened window and invite your teammate. **Always** submit your results choosing this organization.
- Make sure you appear on the leaderboard until the project's final report deadline.

## 4 Interim Report & Progress Meeting (Optional)

Progress meetings will be held for groups that wish to receive detailed feedback on their project progress during the term.

- To participate, groups must upload an interim report in advance that follows the structure of the final report, with a stronger focus on model details and results (See Section 5).
- Groups that choose not to attend a meeting are not required to submit an interim report.
- **Up to 20 bonus points** may be awarded to groups that submit an interim report **and** attend a meeting.
- Details on how to book an appointment, as well as the location, schedule, and duration of the meetings, will be announced later.

Progress meetings will be held in **Week 8** (8–10 April), and the deadline for the interim report is **7 April at 23:59**.

## 5 Project Code & Final Report

You should **write a final report** using the **ACL template**. The report should be **3 or 4 pages** long (except references and appendices). You should:

- Introduce the problem and the dataset,
- Have an extensive literature review,
- Explain your models in detail,
- Analyse and discuss your results.
- Check this for quick summary on how to write a NLP paper.

The deadline for submitting final report and project code is **May 12, 23:59**.

# 6 Demo

As part of the project evaluation, a demo session will be held with the participation of **all** team members.

- Before the demo, make sure your code works from start to finish. Prepare a small sample dataset for situations where the training and testing processes take a long time.
- The demo will take place in **Week 13 & 14** (May 13-15, May 18-19). Details about the place and time will be announced later.

# 7 Presentation

In **Week 14** (May 20), you will be expected to give a 6-minute presentation (+2 minutes Q&A) summarizing your project work in class and answering the questions.

- All presentations will be delivered from a single shared computer; therefore, please upload your presentation to Ninova in advance.
- Time limit is **absolute and you must not exceed it**. You are suggested to practice beforehand.
- Since all groups are working on similar projects, do not spend time explaining the common task or the shared dataset, unless you use additional datasets. Instead, briefly mention them and focus directly on how you approached and implemented the project.
- Clearly explain the specific methods, tools, or techniques you used. Be concise and focus on the most impactful aspects of your process.
- Clearly articulate your main outcomes, discoveries, or solutions. Use visuals like graphs, charts, or images to effectively convey your insights.
- Briefly discuss any significant obstacles you encountered during the project, address limitations of your approach, and explain how you overcame them (if you did).
- Summarize your main takeaways and discuss potential next steps or areas for further development based on your findings.
- Aim for 5-7 slides to effectively convey your message. Each slide should have a clear purpose and not be overly text-heavy. Use visuals and bullet points to highlight key information. Ensure your slides are well-designed and free of typos.
- Your language should be clear, concise, and easy to understand. Maintain eye contact with your audience, project your voice clearly, and speak at a steady pace. Show enthusiasm for your work!

# 8 Grading

- Proposal %5
- Code %30
- Demo %30
- Report %25
- Presentation %10
- Interim Report & Progress Meeting (Bonus/Optional) %20

# 9 Timeline

| Project Announcement | Proposal Submission | Interim Report Submission (Opt.) | Final Report & Code Submission | Demo | Presentation |
|---|---|---|---|---|---|
| 18.02.26 | 03.03.26 | 07.04.26 | 12.05.26 | 13.05.26-19.05.26 | 20.05.24 |

Figure 1: Project Timeline

# 10 Other Considerations

You are also kindly requested to consider the following issues throughout the project:

- Completing the project without any meaningful model development will result in failure of the project (and the course), even if some requirements (such as the proposal and final report) are fulfilled.
- Projects will be evaluated taking into account both the difficulty of the chosen task and the overall complexity and ambition of the proposed approach.
- Use formal **academic** language. Proofread for grammar and clarity.
- Follow ACL-style citations and include a proper references section.
- You are allowed (and encouraged) to use generative AI tools (e.g., ChatGPT, Claude, Copilot) to assist with writing and coding only under the following conditions:
  - Transparency is required. You must **clearly** state in a footnote or a separate section (e.g., Acknowledgments) if and how you used AI tools. For example:

    "*We used ChatGPT (or different LLMs) to help rephrase certain sentences in the Related Work section and to generate an initial draft of the model description, which was then revised and edited by the authors.*"
  - If any generative AI tool was used, the corresponding interaction history must be made publicly available and referenced via a hyperlink in the Acknowledgments or footnote that discloses AI usage.
  - Do not submit unedited AI-generated content. Submissions that are primarily or entirely generated by AI tools without meaningful human input or revision will receive a grade penalty and may be subject to academic integrity review.
  - You are responsible for factual and academic accuracy. Generative models may produce hallucinated or inaccurate claims, especially in literature reviews or citations. You must verify all content and references yourself.
  - Do not use AI to generate fake citations. All references must correspond to real, peer-reviewed publications. Citations fabricated by AI tools will be treated as academic misconduct.
  - The ideas and analysis must be your own. While AI can help with phrasing or summarizing, the core project design, interpretation of results, and critical discussion must reflect your own thinking.
- The project is a group work of 3 people (except Project 1.3). Make sure that the workload is **evenly distributed** among the group members and that they are **actively involved** in every step of the project.
- All final submissions will be made through **only** Ninova and their deadlines are strict. It is recommended that you upload regularly to avoid any problems.
- For any other issues not included in this document, please refer to the ITU Guideline on Ethical Values in University Life.